

chapter 14

fourteen

accounting and financial statements

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LEARNING OBJECTIVES

After reading this chapter, you will be able to:

- LO 14-1** Describe the different uses of accounting information.
- LO 14-2** Demonstrate the accounting process.
- LO 14-3** Examine the various components of an income statement in order to evaluate a firm's "bottom line."

- LO 14-4** Interpret a company's balance sheet to determine its current financial position.
- LO 14-5** Analyze financial statements, using ratio analysis, to evaluate a company's performance.

Although you may cover some of this material in your accounting course, reading this chapter will only strengthen your understanding of accounting. What professors find is that a little duplication and repetition goes a long way in helping the brain retain material. Accounting is the financial “language” that organizations use to record, measure, and interpret all of their financial transactions and records and is very important in business. All businesses—from a small family farm to a giant corporation—use the language of accounting to make sure they track their use of funds, measure profitability,

and budget for future expenditures. Nonbusiness organizations such as charities and governments also use accounting to demonstrate to donors and taxpayers how well they use their funds to meet their stated objectives.

This chapter explores the role of accounting in business and its importance in making business decisions. First, we discuss the uses of accounting information and the accounting process. Then, we briefly look at some simple financial statements and accounting tools that are useful in analyzing organizations worldwide. ■

LO 14-1 Describe the different uses of accounting information.

THE NATURE OF ACCOUNTING

Simply stated, **accounting** is the recording, measurement, and interpretation of financial information. Large numbers of people and institutions, both within and outside businesses, use accounting tools to evaluate organizational operations. The Financial Accounting Standards Board has been setting the principles and standards of financial accounting and reporting in the private sector since 1973. Its mission is to establish and improve standards of financial accounting and reporting for the guidance and education of the public, including issuers, investors, auditors, and users of financial information. However, accounting scandals at the turn of the last century resulted when many accounting firms and businesses failed to abide by generally accepted accounting principles, or GAAP. Consequently, the federal government has taken a greater role in making rules, requirements, and policies for accounting firms and businesses through the Securities and Exchange Commission’s (SEC) Public Company Accounting Oversight Board (PCAOB). For example, the PCAOB has the ability to file a disciplinary order against a firm or individual that either temporarily or permanently prohibits that firm or individual from practicing accounting. To better understand the importance of accounting, we

must first understand who prepares accounting information and how it is used.

Accountants

Many of the functions of accounting are carried out by public or private accountants.

Public Accountants.

Individuals and businesses can hire a **certified public accountant (CPA)**, an individual who has been certified by the state in which he or she practices to provide accounting services ranging from the preparation of financial records and the filing of tax returns to complex audits of corporate financial records. Certification gives a public accountant the right to express, officially, an unbiased opinion regarding the accuracy of the client’s financial statements. Most public accountants are either self-employed or members of large public accounting firms such as Ernst & Young, KPMG, Deloitte, and PricewaterhouseCoopers, together referred to as “the Big Four.” In addition, many CPAs work for one of the second-tier accounting firms that are much smaller than the Big Four. Table 14.1 shows the number of employees at the most prestigious accounting firms. Vault.com uses a weighted ranking system based on survey results to create a score that represents work-life quality issues to reflect the prestige of the firm. Note that not all of the prestigious firms are ranked high in work-life quality.

accounting the recording, measurement, and interpretation of financial information.

certified public accountant (CPA) an individual who has been state certified to provide accounting services ranging from the preparation of financial records and the filing of tax returns to complex audits of corporate financial records.

▼ **TABLE 14.1** Most Prestigious Accounting Firms in the United States.

Rank	Firm	Employees	Location
1	PwC (PricewaterhouseCoopers) LLP	58,133	New York, NY
2	Deloitte LLP	80,000	New York, NY
3	Ernst & Young LLP (EY)	45,000	New York, NY
4	KPMG LLP	31,000	New York, NY
5	Grant Thornton LLP	7,530	Chicago, IL
6	BDO USA LLP	5,395	Chicago, IL
7	RSM US LLP	9,560	Chicago, IL
8	Baker Tilly Virchow Kraus LLP	2,800	Chicago, IL
9	Crowe LLP	4,100	Chicago, IL
10	Moss Adams LLP	2,500	Seattle, WA

Note: U.S. only, does not include international operations.

Source: “Most Prestigious Accounting Firms,” Vault.com, <https://www.vault.com/best-companies-to-work-for/accounting/most-prestigious-accounting-firms> (accessed August 28, 2020).

private accountants

accountants employed by large corporations, government agencies, and other organizations to prepare and analyze their financial statements.

certified management accountants (CMAs)

private accountants who, after rigorous examination, are certified by the Institute of Management Accountants and who have some managerial responsibility.

Although there will always be companies and individual money managers who can successfully hide illegal or misleading accounting practices for a while, eventually they are exposed. After the accounting scandals of Enron and Worldcom in the early 2000s, Congress passed the Sarbanes-Oxley Act, which required firms to be more rigorous in their accounting and reporting practices. Sarbanes-Oxley made accounting firms separate their consulting and auditing businesses and punished corporate executives with potential jail sentences for inaccurate, misleading, or illegal accounting statements. This seemed to reduce the accounting errors among nonfinancial companies,

but declining housing prices exposed some of the questionable practices by banks and mortgage companies. Only five years after the passage of the Sarbanes-Oxley Act, the world experienced a financial crisis starting in 2008—part of which was due to excessive risk taking and inappropriate accounting practices. Many banks failed to understand the true state of their financial health. Banks also developed questionable lending practices and investments based on subprime mortgages made to individuals who had poor credit. When housing prices declined and people suddenly found that they owed more on their mortgages than their homes were worth, they began to default. To prevent a depression, the government intervened and bailed out some of the

United States’ largest banks. Congress passed the Dodd-Frank Act in 2010 to strengthen the oversight of financial institutions. This act gave the Federal Reserve Board the task of implementing the legislation. This legislation limits the types of assets commercial banks can buy; the amount of capital they must maintain; and the use of derivative instruments such as options, futures, and structured investment products. However, at the Economics Club of New York on February 16, 2017, Alan Greenspan, the former chair of the Federal Reserve Board, was quoted as saying the Dodd-Frank Act is one of the worst pieces of legislation he has ever seen.

A growing area for public accountants is *forensic accounting*, which is accounting that is fit for legal review. It involves analyzing financial

documents in search of fraudulent entries or financial misconduct. Functioning as much like detectives as accountants, forensic accountants have been used since the 1930s. Many auditing firms are expanding their forensic or fraud-detection services. Additionally, many forensic accountants root out evidence of “cooked books” for federal agencies like the Federal Bureau of Investigation or the Internal Revenue Service. The Association of Certified Fraud Examiners, which certifies accounting professionals as *certified fraud examiners (CFEs)*, has grown to more than 85,000 members.¹

Private Accountants. Large corporations, government agencies, and other organizations may employ their own **private accountants** to prepare and analyze their financial statements. With titles such as controller, tax accountant, or internal auditor, private accountants are deeply involved in many of the most important financial decisions of the organizations for which they work. Private accountants can be CPAs and may become **certified management accountants (CMAs)** by passing a rigorous examination by the Institute of Management Accountants.

Accounting or Bookkeeping?

The terms *accounting* and *bookkeeping* are often mistakenly used interchangeably. Much narrower and far more mechanical than accounting, bookkeeping is typically limited to the routine, day-to-day recording of business transactions. Bookkeepers are responsible for obtaining and recording the information that accountants require to analyze a firm’s financial position. They generally require less training than accountants. Accountants, on the other hand, usually complete course work beyond their basic four- or five-year college accounting degrees. This additional training allows accountants not only to record financial information



DID YOU KNOW?

Corporate fraud costs are estimated at \$5 trillion annually.^a



Ernst & Young is part of the Big Four, or the four largest international accounting firms. The other three are KPMG, PricewaterhouseCoopers, and Deloitte.

Lars Niki

but to understand, interpret, and even develop the sophisticated accounting systems necessary to classify and analyze complex financial information.

The Uses of Accounting Information

Accountants summarize the information from a firm's business transactions in various financial statements (which we'll look at in a later section of this chapter) for a variety of stakeholders, including managers, investors, creditors, and government agencies. Many business failures may be directly linked to ignorance of the information "hidden" inside these financial statements. Likewise, most business successes can be traced to informed managers who understand the consequences of their decisions. While maintaining and even increasing short-run profits is desirable, the failure to plan sufficiently for the future can easily lead an otherwise successful company to insolvency and bankruptcy court.

Basically, managers and owners use financial statements (1) to aid in internal planning and control and (2) for external purposes such as reporting to the Internal Revenue Service, stockholders, creditors, customers, employees, and other interested parties. Figure 14.1 shows some of the users of the accounting information generated by organizations and other stakeholders.

Internal Uses. Managerial accounting

refers to the internal use of accounting statements by managers in planning and directing the organization's activities. Perhaps management's greatest single concern is **cash flow**, the movement of money through an organization over a daily, weekly, monthly, or yearly basis. Obviously, for any business to succeed, it needs to generate enough cash to pay its bills as they fall due. However, it is not at all unusual for highly successful and rapidly growing companies to struggle to make payments to employees, suppliers, and lenders because of an inadequate cash flow. One common reason for a so-called cash crunch, or shortfall, is poor managerial planning.

Managerial accountants also help prepare an organization's **budget**, an internal financial plan that forecasts expenses and income over a set period of time. It is not unusual for an organization to prepare separate daily, weekly, monthly, and yearly budgets. Think of a budget as a financial map, showing how the company expects to move from Point A to Point B over a specific period of time. While most companies prepare *master budgets* for the entire firm, many also prepare budgets for smaller segments of the organization such as divisions, departments, product lines, or projects. "Top-down" master budgets begin at the upper management level and filter down to the individual department level, while "bottom-up" budgets start at the department or project level

managerial accounting

the internal use of accounting statements by managers in planning and directing the organization's activities.

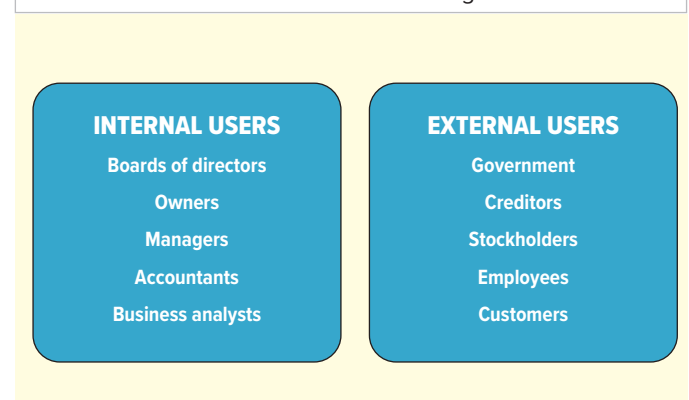
cash flow

the movement of money through an organization over a daily, weekly, monthly, or yearly basis.

budget

an internal financial plan that forecasts expenses and income over a set period of time.

FIGURE 14.1 The Users of Accounting Information



HOLDING ERNST & YOUNG ACCOUNTABLE

From its small beginnings as separate accounting firms, Ernst & Young (EY) has evolved into one of the four largest firms in the accounting world. Currently, EY has more than 280,000 global employees, global revenues of \$36.4 billion, and more than 700 offices worldwide.

Alwin Ernst and Arthur Young originally founded two separate accounting firms more than a century ago. Young formed his accounting firm, Arthur Young, in 1906, and Ernst began Ernst & Ernst (later Ernst & Whinney) in 1903. These firms grew over the years and became part of the big eight accounting firms in the 1970s and 1980s. In 1989, the two firms merged to create EY.

Today there are four major segments that make up EY's structure: advisory, assurance,

tax, and transaction advisory services. Assurance or auditing is one of the major building blocks of effective capital markets as it provides confidence and integrity in companies' financial statements. EY's advisory services have surpassed its tax business to make up 28 percent of total revenue. Transaction advisory services is another rapidly growing segment similar to advisory services but focuses on strategies implemented on a transaction-by-transaction basis. Though EY recently fell in the Vault Accounting 50 rankings of best accounting firms to work for, these business segments offer many opportunities for aspiring accountants which—when combined with employee benefits like four months of paid time off for parents—have contributed to EY's placement

on *Fortune's* "100 Best Companies to Work For" for two decades.^b

Critical Thinking Questions

1. Why are the assurance services that Ernst & Young provides so important?
2. Ernst & Young offers assurance services and advisory services. What limitations did Sarbanes-Oxley place on accounting firms regarding how they offer these services?
3. Considering the accounting scandals of the early 20th century, do you believe it is advisable to have so much accounting power consolidated in four large accounting firms? Why or why not?

annual report

summary of a firm's financial information, products, and growth plans for owners and potential investors.

and are combined at the chief executive's office. Generally, the larger and more rapidly growing an organization, the greater will be the likelihood that it will build its master budget from the ground up.

Regardless of focus, the principal

value of a budget lies in its breakdown of cash inflows and outflows. Expected operating expenses (cash outflows such as wages, materials costs, and taxes) and operating revenues (cash inflows in the form of payments from customers) over a set period of time are carefully forecast and subsequently compared with actual results. Deviations between the two serve as a "trip wire" or "feedback loop" to launch more detailed financial analyses in an effort to pinpoint trouble spots and opportunities.

External Uses. Managers also use accounting statements to report the business's financial performance to outsiders. Such statements are used for filing income taxes, obtaining credit from lenders, and reporting results to the firm's stockholders. They become the basis for the information provided in the official corporate **annual report**,

UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

FORM 10-K

☒ ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934
For the fiscal year ended January 26, 2020

OR

☐ TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934

Commission file number: 0-23985



NVIDIA CORPORATION

(Exact name of registrant as specified in its charter)

Delaware
(State or other jurisdiction of
Incorporation or Organization)

94-3177549
(I.R.S. Employer
Identification No.)

2788 San Tomas
Expressway Santa Clara,
California 95051 (408) 486-
2000

(Address, including zip code, and telephone number, including area code, of principal executive offices) Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol(s)	Name of each exchange on which registered
Common Stock, \$0.001 par value per share	NVDA	The Nasdaq Global Select Market

Securities registered pursuant to Section 12(g) of the Act:
None

The annual report is a summary of the firm's financial information, products, and growth plans for owners and potential investors. Many investors look at a firm's annual report to determine how well the company is doing financially.

Source: U.S. Securities and Exchange Commission.

a summary of the firm's financial information, products, and growth plans for owners and potential investors. While frequently presented between slick, glossy covers, the single most important component of an annual report is the signature of a certified public accountant attesting that the required financial statements are an accurate reflection of the underlying financial condition of the firm. Financial statements meeting these conditions are termed *audited*. The primary external users of audited accounting information are government agencies, stockholders and potential investors, lenders, suppliers, and employees.

During the global financial crisis, it was discovered that Greece had been engaging in deceptive accounting practices using financial techniques to hide massive amounts of debt from its public balance sheets. Eventually, the markets figured out the country might not be able to pay off its creditors. The European Union and the International Monetary Fund bailed out Greece with loans and credit relief, but tied to this was the message to “get your financial house in order.”

Many states have huge pension fund deficits and some have been borrowing from pension funds to help balance their budgets which only compounds the problem. To make matters worse, the COVID-19 (coronavirus) pandemic in 2020 caused major financial difficulties for states as they had to cover unemployment benefits and fight the virus with unexpected medical costs. States like New York, Illinois, Connecticut, New Jersey, and Louisiana were hardest hit because of their high population density cities like Chicago and New Orleans.

Financial statements evaluate the return on stockholders' investments and the overall quality of the firm's management team. As a result, poor performance, as documented in the financial statements, often results in changes in top management.

Potential investors study the financial statements in a firm's annual report to determine whether the company meets its investment requirements and whether the returns from a given firm are likely to compare favorably with other similar companies. NVIDIA's annual financial statement submitted to the Securities and Exchange Commission, called Form 10-K, can be seen on the previous page.

Banks and other lenders look at financial statements to determine a company's ability to meet current and future debt obligations if a loan or credit is granted. To determine this ability, a short-term lender examines a firm's cash flow to assess its ability to repay a loan quickly with cash generated from sales. A long-term lender is more interested in the company's profitability and indebtedness to other lenders. In the advertisement, HSBC highlights its insights into global business and modern trade through its Navigator survey. HSBC, based in London, is the seventh largest bank in the world and the largest bank in Europe.³ HSBC offers loans for everything from personal use to global business. Evaluating financial statements is an important step in the lending process for banks like HSBC.

Labor unions and employees use financial statements to establish reasonable expectations for salary and other benefit

As one of the biggest banks in the world, HSBC specializes in loans, mortgages, savings, investments, and credit cards. The data provided can be used in financial statements.

Source: HSBC

requests. Just as firms experiencing record profits are likely to face added pressure to increase employee wages, so too are employees unlikely to grant employers wage and benefit concessions without considerable evidence of financial distress.

LO 14-2 Demonstrate the accounting process.

THE ACCOUNTING PROCESS

Many view accounting as a primary business language. It is of little use, however, unless you know how to “speak” it. Fortunately, the fundamentals—the accounting equation and the double-entry bookkeeping system—are not difficult to learn. These two concepts serve as the starting point for all currently accepted accounting principles.

ARTIFICIAL INTELLIGENCE TRANSFORMS ACCOUNTING INTELLIGENCE

Artificial intelligence (AI) is a type of technology that allows machines to perform decision-based tasks previously left to humans. To some accountants, the words “artificial intelligence” mean goodbye jobs. Kai Fu Lee, an AI expert and former Google executive, believes 40 percent of all jobs will be replaced by automation in the next two decades.

On the other hand, AI is currently being used to supplement accountants, not replace them. The director of accounting firm Raedan believes AI is the next step in automation and efficiency provided by cloud software. He believes AI will increase time savings, reduce errors, and aid compliance. In turn, this will allow accountants to spend time on more meaningful tasks, while AI can cover tasks like data entry.

AI may also increase employee satisfaction, empowering accountants to focus more on tasks they find enjoyable. Especially at the Big Four accounting firms, retention is difficult. Employees become exhausted from the amount of work and number of hours they have to put in. If AI can be used to automate some of this work, accountants would suffer less from burnout.

AI has so many advantages for businesses that many are adopting their own systems. For instance, PricewaterhouseCoopers has developed its own AI system called GL.ai. This system identifies anomalies in its clients’ general ledger systems. According to a recent survey of 3,000+ business executives, more than 80 percent of executives believe AI leads to a competitive advantage, and 79 percent believe

it will increase their company’s productivity. AI will disrupt jobs and businesses of all kinds. However, professionals should not be scared to jump on the AI train because it is likely that AI will disrupt businesses for the better.^c

Critical Thinking Questions

1. Describe some of the advantages of AI to the accounting industry.
2. GL.ai, the AI system adopted by PricewaterhouseCoopers, helps identify anomalies in its clients’ general ledger systems. Why do you think this is important?
3. If AI can take over basic accounting tasks such as data entry, do you think this will improve the services accountants offer to their customers? Why or why not?

assets a firm’s economic resources, or items of value that it owns, such as cash, inventory, land, equipment, buildings, and other tangible and intangible things.

liabilities debts that a firm owes to others.

owners’ equity equals assets minus liabilities and reflects historical values.

accounting equation assets equal liabilities plus owners’ equity.

The Accounting Equation

Accountants are concerned with reporting an organization’s assets, liabilities, and owners’ equity. To help illustrate these concepts, consider a hypothetical floral shop called Anna’s Flowers, owned by Anna Rodriguez. A firm’s economic resources, or items of value that it owns, represent its **assets**—cash, inventory, land, equipment, buildings, and other tangible and intangible things. The assets of Anna’s Flowers include counters, refrigerated display cases, flowers, decorations,

vases, cards, and other gifts, as well as something known as “goodwill,” which in this case is Anna’s reputation for preparing and delivering beautiful floral arrangements on a timely basis. **Liabilities**, on the other hand, are debts the firm owes to others. Among the liabilities of Anna’s Flowers are a loan from the Small Business Administration and money owed to flower suppliers and other creditors for items purchased. The **owners’ equity** category contains all of the money that has ever been contributed to the company that never has to be paid back. The funds can come from investors who have given money or assets

to the company, or it can come from past profitable operations. In the case of Anna’s Flowers, if Anna were to sell off, or liquidate, her business, any money left over after selling all the shop’s assets and paying off its liabilities would comprise her owners’ equity. The relationship among assets, liabilities, and owners’ equity is a fundamental concept in accounting and is known as the **accounting equation**:

$$\text{Assets} = \text{Liabilities} + \text{Owners' equity}$$



The owners’ equity portion of this florist’s balance sheet includes the money she has put into the firm.

Jose Luis Pelaez Inc/Blend Images LLC

Double-Entry Bookkeeping

Double-entry bookkeeping is a system of recording and classifying business transactions in separate accounts in order to maintain the balance of the accounting equation. Returning to Anna’s Flowers, suppose Anna buys \$325 worth of roses on credit from the Antique Rose Emporium to fill a wedding order. When she records this transaction, she will list the \$325 as a liability or a debt to a supplier. At the same time, however, she will also record \$325 worth of roses as an asset in an account known as “inventory.” Because the assets and liabilities are on different sides of the accounting equation, Anna’s accounts increase in total size (by \$325) but remain in balance:

Assets = Liabilities + Owners’ equity
\$325 = \$325

Thus, to keep the accounting equation in balance, each business transaction must be recorded in two separate accounts.

In the final analysis, all business transactions are classified as assets, liabilities, or owners’ equity. However, most organizations further break down these three accounts to provide more specific information about a transaction. For example, assets may be broken down into specific categories such as cash, inventory, and equipment, while liabilities may include bank loans, supplier credit, and other debts.

Figure 14.2 shows how Anna used the double-entry bookkeeping system to account for all of the transactions that took place in her first month of business. These transactions include her initial

investment of \$2,500, the loan from the Small Business Administration, purchases of equipment and inventory, and the purchase of roses on credit. In her first month of business, Anna generated revenues of \$2,000 by selling \$1,500 worth of inventory. Thus, she deducts, or (in accounting notation that is appropriate for assets) *credits*, \$1,500 from inventory and adds, or *debits*, \$2,000 to the cash account. The difference between Anna’s \$2,000 cash inflow and her \$1,500 outflow is represented by a credit to owners’ equity because it is money that belongs to her as the owner of the flower shop.

double-entry bookkeeping a system of recording and classifying business transactions that maintains the balance of the accounting equation.

accounting cycle the four-step procedure of an accounting system: examining source documents, recording transactions in an accounting journal, posting recorded transactions, and preparing financial statements.

The Accounting Cycle

In any accounting system, financial data typically pass through a four-step procedure sometimes called the **accounting cycle**. The steps include examining source documents, recording transactions in an accounting journal, posting recorded transactions, and preparing financial statements. Figure 14.3 shows how Anna works through them. Traditionally, all of these steps were performed using paper, pencils, and erasers (lots of erasers!), but today, the process is often fully computerized.

▼ **FIGURE 14.2** The Accounting Equation and Double-Entry Bookkeeping for Anna’s Flowers

	Assets			=		Liabilities	+	Owners’ Equity
	Cash	Equipment	Inventory			Debts to suppliers	Loans	Equity
Cash invested by Anna	\$2,500.00							\$2,500.00
Loan from SBA	\$5,000.00						\$5,000.00	
Purchase of furnishings	−\$3,000.00	\$3,000.00						
Purchase of inventory	−\$2,000.00		\$2,000.00					
Purchase of roses			\$325.00			\$325.00		
First month sales	\$2,000.00		−\$1,500.00					\$500.00
Totals	\$4,500.00	\$3,000.00	\$825.00			\$325.00	\$5,000.00	\$3,000.00
	\$8,325		=			\$5,325	+	\$3,000
	\$8,325 Assets			=		\$5,325 (Liabilities + Owners’ Equity)		

Step 1:
Source documents
show that a
transaction
took place.

Receipt Anna's Flowers	
July 7 Wedding floral arrangements	\$500.00
Consultation services	250.00

Step 2:
The transaction
is recorded
in the journal.

Assets	
Cash	
July 7 Brown wedding	\$750.00

Step 3:
The transaction
is posted to the
general ledger
under the
appropriate
account (asset,
liability, or some
further breakdown
of these main
accounts).

Date		Explanation	PR	Debit	Credit	Balance	
						Debit	Credit
2021							
July	1		1	2,000		2,000	
	3		1		1,250		1,250
	4		1				
	7	Brown wedding	1		750		750
	14		1				

Step 4: At the end
of the accounting
period, the ledger
is used to prepare
the firm's financial
statements.

Anna's Flowers Income Statement December 31, 2021		Anna's Flowers Balance Sheet December 31, 2021		Anna's Flowers Annual Budget for 2021			
Revenues:		<u>Assets</u>		Sales Consulting Total			
Net sales	\$123,850	Current assets:		January	10,500	4,500	15,000
Consulting	73,850	Cash \$17,850		February	10,000	5,500	15,500
Total revenues	\$197,700	Accounts receivable 10,200		March	10,800	5,700	16,500
Expenses:		Merch. Inventory 8,750		April	10,100	6,050	16,150
Cost of goods sold	\$72,600	Tot. assets \$36,800		May	12,000	6,000	18,000
Selling expenses	37,700	Property and Equipment		June	12,100	6,250	18,350
General and admin.	18,400	Equipment 11,050		July	13,000	6,600	19,600
Other expenses	5,600	Office building 73,850		August	9,950	6,000	15,950
Total expenses	134,300	Tot. prop. & equip. 84,900		September	9,700	6,200	15,900
Net income	\$63,400	Total assets \$121,700		October	9,900	7,000	16,900
		<u>Liabilities and Owner's Equity</u>		November	8,500	7,150	15,650
		Current liabilities		December	7,300	6,900	14,200
		Accounts payable \$12,600		Annual	\$123,850	\$73,850	\$197,700
		Tot. cur. liabilities 12,600					
		Long-term liabilities					
		Mortgage payable 23,600					
		Total liabilities 36,200					
		Owner's equity:					
		Anna Rodriguez, capital \$85,500					
		Tot. liabilities and owners' equity \$121,700					

Anna's Flowers

Step One: Examine Source Documents. Like all good managers, Anna Rodriguez begins the accounting cycle by gathering and examining source documents—checks, credit card receipts, sales slips, and other related evidence concerning specific transactions.

Step Two: Record Transactions. Next, Anna records each financial transaction in a **journal**, which is basically just a time-ordered list of account transactions. While most businesses keep a general journal in which all transactions are recorded, some classify transactions into specialized journals for specific types of transaction accounts.

Step Three: Post Transactions. Anna next transfers the information from her journal into a **ledger**, a book or computer program with separate files for each account. This process is known as *posting*. At the end of the accounting period (usually yearly, but occasionally quarterly or monthly), Anna prepares a *trial balance*, a summary of the balances of all the accounts in the general ledger. If, upon totaling, the trial balance doesn't balance (that is, the accounting equation is not in balance), Anna or her accountant must look for mistakes (typically an error in one or more of the ledger entries) and correct them. If the trial balance is correct, the accountant can then begin to prepare the financial statements.

Step Four: Prepare Financial Statements. The information from the trial balance is also used to prepare the company's financial statements. In the case of public corporations and certain other organizations, a CPA must *attest*, or certify, that the organization followed generally accepted accounting principles in preparing the financial statements. When these statements have been completed, the organization's books are "closed," and the accounting cycle begins anew for the next accounting period.

FINANCIAL STATEMENTS

The end result of the accounting process is a series of financial statements. The income statement, the balance sheet, and the statement of cash flows are the best-known examples of financial statements. They are provided to stockholders and potential investors in a firm's annual report as well as to other relevant outsiders such as creditors, government agencies, and the Internal Revenue Service.

It is important to recognize that not all financial statements follow precisely the same format. The fact that different organizations generate income in different ways suggests that when it comes to financial statements, one size definitely does not fit all. Manufacturing firms, service providers, and nonprofit organizations each use a different set of accounting principles

or rules upon which the public accounting profession has agreed. As we have already mentioned, these are sometimes referred to as *generally accepted accounting principles (GAAP)*. Each country has a different set of rules that the businesses within that country are required to use for their accounting process and financial statements. However, a number of countries have adopted a standard set of accounting principles known as International Financial Reporting Standards. The United States has discussed adopting these standards to create a more standardized system of reporting for global investors. Moreover, as is the case in many other disciplines, certain concepts have more than one name. For example, *sales* and *revenues* are often interchanged, as are *profits*, *income*, and *earnings*. Table 14.2 lists a few common equivalent terms that should help you decipher their meaning in accounting statements.

journal a time-ordered list of account transactions.

ledger a book or computer file with separate sections for each account.

income statement a financial report that shows an organization's profitability over a period of time—month, quarter, or year.

LO 14-3 Examine the various components of an income statement in order to evaluate a firm's "bottom line."

The Income Statement

The question, "What's the bottom line?" derives from the income statement, where the bottom line shows the overall profit or loss of the company after taxes. Thus, the **income statement** is a financial report that shows an organization's profitability over

▼ **TABLE 14.2** Equivalent Terms in Accounting

Term	Equivalent Term
Revenues	Sales
	Goods or services sold
Gross profit	Gross income
	Gross earnings
Operating income	Operating profit
	Earnings before interest and taxes (EBIT)
	Income before interest and taxes (IBIT)
Income before taxes (IBT)	Earnings before taxes (EBT)
	Profit before taxes (PBT)
Net income (NI)	Earnings after taxes (EAT)
	Profit after taxes (PAT)
Income available to common stockholders	Earnings available to common stockholders

revenue the total amount of money received from the sale of goods or services, as well as from related business activities.

cost of goods sold the amount of money a firm spent to buy or produce the products it sold during the period to which the income statement applies.

a period of time, be that a month, quarter, or year. By its very design, the income statement offers one of the clearest possible pictures of the company’s overall revenues and the costs incurred in generating those revenues. Other names for the income statement include profit and loss (P&L) statement or operating statement. A sample income statement in word form with line-by-line explanations is presented in Table 14.3, while

Table 14.4 presents recent income statements of Nvidia. The income statement indicates the firm’s profitability or income (the bottom line), which is derived by subtracting the firm’s expenses from its revenues.

Revenue. **Revenue** is the total amount of money received (or promised) from the sale of goods or services, as well as from other business activities such as the rental of property and investments. Nonbusiness entities typically obtain revenues through donations from individuals and/or grants from governments and private foundations. One of the controversies in accounting has been when a business should recognize revenue. For instance, should an organization book revenue during a project or after the project is completed? Differences in revenue recognition have caused similar organizations to book different accounting results. Generally accepted practice is that firms should book revenue when “it satisfies a performance obligation by transferring a promised good or service to a customer.”⁴

For most manufacturing and retail concerns, the next major item included in the income statement is the **cost of goods sold**, the amount of money the firm spent (or promised to

▼ **TABLE 14.3** Sample Income Statement

Company Name for the Year Ended December 31	
Revenues (sales)	Total dollar amount of products sold (includes income from other business services such as rental-lease income and interest income).
Less: Cost of goods sold	The cost of producing the goods and services, including the cost of labor and raw materials as well as other expenses associated with production.
Gross profit	The income available after paying all expenses of production.
Less: Selling and administrative expense	The cost of promoting, advertising, and selling products as well as the overhead costs of managing the company. This includes the cost of management and corporate staff. One noncash expense included in this category is depreciation, which approximates the decline in the value of plant and equipment assets due to use over time. In most accounting statements, depreciation is not separated from selling and administrative expenses. However, financial analysts usually create statements that include this expense.
Income before interest and taxes (operating income or EBIT)	This line represents all income left over after operating expenses have been deducted. This is sometimes referred to as operating income because it represents all income after the expenses of operations have been accounted for. Occasionally, this is referred to as EBIT, or earnings before interest and taxes.
Less: Interest expense	Interest expense arises as a cost of borrowing money. This is a financial expense rather than an operating expense and is listed separately. As the amount of debt and the cost of debt increase, so will the interest expense. This covers the cost of both short-term and long-term borrowing.
Income before taxes (earnings before taxes—EBT)	The firm will pay a tax on this amount. This is what is left of revenues after subtracting all operating costs, depreciation costs, and interest costs.
Less: Taxes	The tax rate is specified in the federal tax code.
Net income	This is the amount of income left after taxes. The firm may decide to retain all or a portion of the income for reinvestment in new assets. Whatever it decides not to keep will usually be paid out in dividends to its stockholders.
Less: Preferred dividends	If the company has preferred stockholders, they are first in line for dividends. That is one reason their stock is called “preferred.”
Income to common stockholders	This is the income left for the common stockholders. If the company has a good year, there may be a lot of income available for dividends. If the company has a bad year, income could be negative. The common stockholders are the ultimate owners and risk takers. They have the potential for very high or very poor returns because they get whatever is left after all other expenses.
Earnings per share	Earnings per share is found by taking the income available to the common stockholders and dividing by the number of shares of common stock outstanding. This is income generated by the company for each share of common stock.

gross income

revenues minus the cost of goods sold required to generate the revenues.

profit the difference between what it costs to make and sell a product and what a customer pays for it.

expenses the costs incurred in the day-to-day operations of an organization.

depreciation the process of spreading the costs of long-lived assets such as buildings and equipment over the total number of accounting periods in which they are expected to be used.

spend) to buy and/or produce the products it sold during the accounting period. This figure may be calculated as follows:

$$\text{Cost of goods sold} = \text{Beginning inventory} + \text{Interim purchases} - \text{Ending inventory}$$

Let's say that Anna's Flowers began an accounting period with an inventory of goods for which it paid \$5,000. During the period, Anna bought another \$4,000 worth of goods, giving the shop a total inventory available for sale of \$9,000. If, at the end of the accounting period, Anna's inventory was worth \$5,500, the cost of goods sold during the period would have been \$3,500

(\$5,000 + \$4,000 - \$5,500 = \$3,500). If Anna had total revenues of \$10,000 over the same period of time, subtracting the cost of goods sold (\$3,500) from the total revenues of \$10,000 yields the store's **gross income** or **profit** (revenues minus the cost of goods sold required to generate the revenues): \$6,500. In the case of Nvidia shown in Table 14.4, the company bundles "the cost of goods sold" with sales, general and administrative costs, so it is not an item they wish to highlight.

Expenses. **Expenses** are the costs incurred in the day-to-day operations of an organization. Three common expense accounts shown on income statements are (1) selling, general, and administrative expenses; (2) research, development, and engineering expenses; and (3) interest expenses (remember that the costs

directly attributable to selling goods or services are included in the cost of goods sold). Selling expenses include advertising and sales salaries. General and administrative expenses include salaries of executives and their staff and the costs of owning and maintaining the general office. Research and development costs include scientific, engineering, and marketing personnel and the equipment and information used to design and build prototypes and samples. Interest expenses include the direct costs of borrowing money.

The number and type of expense accounts vary from organization to organization. Included in the general and administrative category is a special type of expense known as **depreciation**, the process of spreading the costs of long-lived assets such as buildings and equipment over the total number of accounting periods in which they are expected to be used. Consider a manufacturer that purchases a \$100,000 machine expected to last about 10 years. Rather than showing an expense of \$100,000 in the first year and no expense for that equipment over the next nine years, the manufacturer is allowed to report depreciation expenses of \$10,000 per year in each of the next 10 years because that better matches the cost of the machine to the years the machine is used. Each time this depreciation is "written off" as an expense, the book value of the machine is also reduced by \$10,000.

▼ **TABLE 14.4** Nvidia Corporation and Subsidiaries Consolidated Statements of Income (in millions, except per share data)

	Year Ended		
	January 26, 2020	January 27, 2019	January 28, 2018
Revenue	\$10,918	\$11,716	\$9,714
Cost of revenue	4,150	4,545	3,892
Gross profit	6,768	7,171	5,822
Operating expenses			
Research and development	2,829	2,376	1,797
Sales, general and administrative	1,093	991	815
Total operating expenses	3,922	3,367	2,612
Income from operations	2,846	3,804	3,210
Interest income	178	136	69
Interest expense	(52)	(58)	(61)
Other, net	(2)	14	(22)
Total other income (expense)	124	92	(14)
Income before income tax	2,970	3,896	3,196
Income tax expense (benefit)	174	(245)	149
Net income	\$ 2,796	\$ 4,141	\$3,047
Net income per share			
Basic	\$ 4.59	\$ 6.81	\$ 5.09
Diluted	\$ 4.52	\$ 6.63	\$ 4.82
Weighted average shares used in per share computation:			
Basic	609	608	599
Diluted	618	625	632

net income the total profit (or loss) after all expenses, including taxes, have been deducted from revenue; also called net earnings.

balance sheet

a “snapshot” of an organization’s financial position at a given moment.

The fact that the equipment has a zero value on the firm’s balance sheet when it is fully depreciated (in this case, after 10 years) does not necessarily mean that it can no longer be used or is economically worthless. Indeed, in some industries, machines used every day have been reported as having no book value whatsoever for more than 30 years.

Net Income. **Net income** (or net earnings) is the total profit (or loss) after all expenses, including taxes, have been deducted from revenue. Generally, accountants divide profits into individual sections such as operating income and earnings before interest and taxes. Like most companies, Nvidia presents not only the current year’s results, but also the previous two years’ income statements to permit comparison of performance from one period to another.

Net income is very sensitive to the corporate tax rate, and in 2017, Congress implemented the Tax Cut and Jobs Act which became effective in 2018. The United States switched to a territorial tax system consistent with most developed countries so that income is taxed in the country where it was earned. For example, income generated in France is taxed at the French tax rate and not the U.S. rate. The law also reduced the statutory rate on corporate income from 35 percent to 21 percent which made it more comparable to rates in most developed countries.

Temporary Nature of the Income Statement

Accounts. Companies record their operational activities in the revenue and expense accounts during an accounting period. Gross profit, earnings before interest and taxes, and net income are the results of calculations made from the revenues and expenses accounts; they are not actual accounts. At the end of each accounting period, the dollar amounts in all the revenue and expense accounts are moved into an account called “Retained Earnings,” one of the owners’ equity accounts. Revenues increase owners’ equity, while expenses decrease it. The resulting change in the owners’ equity account is exactly equal to the net income. This shifting of dollar values from the revenue and expense accounts allows the firm to begin the next accounting period with zero balances in those accounts. Zeroing out the balances enables a company to count how much it has sold and how many expenses have been incurred during a period of time. The basic accounting equation ($\text{Assets} = \text{Liabilities} + \text{Owners' equity}$) will not balance until the revenue and expense account balances have been moved or “closed out” to the owners’ equity account.

One final note about income statements: You may remember that corporations may choose to make cash payments called



Walmart is the largest company in the United States, with more than \$524 billion in revenue.

Quality HD/Shutterstock

dividends to shareholders out of their net earnings. When a corporation elects to pay dividends, it decreases the cash account (in the assets category of the balance sheet) as well as a capital account (in the owners’ equity category of the balance sheet). During any period of time, the owners’ equity account may change because of the sale of stock (or contributions/withdrawals by owners), the net income or loss, or the dividends paid.

LO 14-4 Interpret a company’s balance sheet to determine its current financial position.

The Balance Sheet

The second basic financial statement is the **balance sheet**, which presents a “snapshot” of an organization’s financial position at a given moment. As such, the balance sheet indicates what the organization owns or controls and the various sources of the funds used to pay for these assets, such as bank debt or owners’ equity.

The balance sheet takes its name from its reliance on the accounting equation: $\text{Assets} = \text{Liabilities} + \text{Owners' equity}$. Assets *must* equal liabilities plus owners’ equity. Table 14.5 provides a sample balance sheet in word form with line-by-line explanations. Unlike the income statement, the balance sheet does not represent the result of transactions completed over a specified accounting period. Instead, the balance sheet is, by definition, an accumulation of all financial transactions conducted by an organization since its founding. Following long-established traditions, items on the balance sheet are listed on the basis of their original cost less accumulated depreciation, rather than their present values.

▼ **TABLE 14.5** Sample Balance Sheet

Typical Company	December 31
Assets	This is the major category for all physical, monetary, or intangible goods that have some dollar value.
Current assets	Assets that are either cash or are expected to be turned into cash within the next 12 months.
Cash	Cash or checking accounts.
Marketable securities	Short-term investments in securities that can be converted to cash quickly (liquid assets).
Accounts receivable	Cash due from customers in payment for goods received. These arise from sales made on credit.
Inventory	Finished goods ready for sale, goods in the process of being finished, or raw materials used in the production of goods.
Prepaid expense	A future expense item that has already been paid, such as insurance premiums or rent.
Total current assets	The sum of the above accounts.
Fixed assets	Assets that are long term in nature and have a minimum life expectancy that exceeds one year.
Investments	Assets held as investments rather than assets owned for the production process. Most often, the assets include small ownership interests in other companies.
Gross property, plant, and equipment	Land, buildings, and other fixed assets listed at original cost.
Less: Accumulated depreciation	The accumulated expense deductions applied to all plant and equipment over their life. Land may not be depreciated. The total amount represents, in general, the decline in value as equipment gets older and wears out. The maximum amount that can be deducted is set by the U.S. Federal Tax Code and varies by type of asset.
Net property, plant, and equipment	Gross property, plant, and equipment minus the accumulated depreciation. This amount reflects the book value of the fixed assets and not their value if sold.
Other assets	Any other asset that is long term and does not fit into the preceding categories. It could be patents or trademarks.
Total fixed assets	The sum of the above accounts.
Total assets	The sum of all the asset values.
Liabilities and stockholders' equity	This is the major category. Liabilities refer to all indebtedness and loans of both a long-term and short-term nature. Stockholders' equity refers to all money that has been contributed to the company over the life of the firm by the owners.
Current liabilities	Short-term debt expected to be paid off within the next 12 months.
Accounts payable	Money owed to suppliers for goods ordered. Firms usually have between 30 and 90 days to pay this account, depending on industry norms.
Wages payable	Money owed to employees for hours worked or salary. If workers receive checks every two weeks, the amount owed should be no more than two weeks' pay.
Taxes payable	Firms are required to pay corporate taxes quarterly. This refers to taxes owed based on earnings estimates for the quarter.
Notes payable	Short-term loans from banks or other lenders.
Other current liabilities	The other short-term debts that do not fit into the preceding categories.
Total current liabilities	The sum of the preceding accounts.
Long-term liabilities	All long-term debt that will not be paid off in the next 12 months.
Long-term debt	Loans of more than one year from banks, pension funds, insurance companies, or other lenders. These loans often take the form of bonds, which are securities that may be bought and sold in bond markets.
Deferred income taxes	This is a liability owed to the government but not due within one year.
Other long-term liabilities	Any other long-term debt that does not fit the preceding two categories.
Total long-term liabilities	The sum of the preceding accounts.
Total liabilities	The sum of all liability values.

(Continued)

▼ **TABLE 14.5** Continued

Typical Company	December 31
Stockholders' equity	The following three categories are the owners' investment in the company.
Common stock	The tangible evidence of ownership is a security called common stock. The par value is stated value and does not indicate the company's worth.
Capital in excess of par (a.k.a. contributed capital)	When shares of stock were sold to the owners, they were recorded at the price at the time of the original sale. If the price paid was \$10 per share, the extra \$9 per share would show up in this account at 100,000 shares times \$9 per share, or \$900,000.
Retained earnings	The total amount of earnings the company has made during its life and not paid out to its stockholders as dividends. This account represents the owners' reinvestment of earnings into company assets rather than payments of cash dividends. This account does not represent cash.
Total stockholders' equity	This is the sum of the preceding equity accounts representing the owners' total investment in the company.
Total liabilities and stockholders' equity	The total short-term and long-term debt of the company plus the owners' total investment. This combined amount <i>must</i> equal total assets.

current assets assets that are used or converted into cash within the course of a calendar year; also called short-term assets.

accounts receivable money owed a company by its clients or customers who have promised to pay for the products at a later date.

current liabilities a firm's financial obligations to short-term creditors, which must be repaid within one year.

accounts payable the amount a company owes to suppliers for goods and services purchased with credit.

accrued expenses all unpaid financial obligations incurred by an organization.

Balance sheets are often presented in two different formats. The traditional balance sheet format placed the organization's assets on the left side and its liabilities and owners' equity on the right. More recently, a vertical format, with assets on top followed by liabilities and owners' equity, has gained wide acceptance. Nvidia's balance sheet is presented in Table 14.6. In the sections that follow, we'll briefly describe the basic items found on the balance sheet; we'll take a closer look at a number of these in the "Financial Management and Securities Markets" chapter.

Assets. All asset accounts are listed in descending order of *liquidity*—that is, how quickly each could be turned into cash. **Current assets**, also called short-term assets, are those that are used or converted into cash within the course of a calendar year. Cash is followed by temporary investments, accounts receivable, and inventory, in that order. **Accounts**

receivable refers to money owed the company by its clients or customers who have promised to pay for the products at a later date. Accounts receivable usually includes an allowance for bad debts that management does not expect to collect. The bad-debts adjustment is normally based on historical collections experience and is deducted from the accounts receivable balance to present a more realistic view of the payments likely to be received in the future, called net receivables. Inventory may be

held in the form of raw materials, work-in-progress, or finished goods ready for delivery.

Long-term or fixed assets represent a commitment of organizational funds of at least one year. Items classified as fixed include long-term investments, such as plants and equipment, and intangible assets, such as corporate "goodwill," or reputation, as well as patents and trademarks.

Liabilities. As seen in the accounting equation, total assets must be financed either through borrowing (liabilities) or through owner investments (owners' equity). **Current liabilities** include a firm's financial obligations to short-term creditors, which must be repaid within one year, while long-term liabilities have longer repayment terms. **Accounts payable** represents amounts owed to suppliers for goods and services purchased with credit. For example, if you buy gas with a Shell credit card, the purchase represents an account payable for you (and an account receivable for Shell). Other liabilities include wages earned by employees but not yet paid and taxes owed to the government. Occasionally, these accounts are consolidated into an **accrued expenses** account, representing all unpaid financial obligations incurred by the organization.

Owners' Equity. Owners' equity includes the owners' contributions to the organization along with income earned by the organization and retained to finance continued growth and development. If the organization were to sell off all of its assets and pay off all of its liabilities, any remaining funds would belong to the owners. Not surprisingly, the accounts listed as owners' equity on a balance sheet may differ dramatically from company to company. Corporations sell stock to investors, who then become the owners of the firm. Many corporations issue two, three, or even more different classes of common and preferred stock, each with different dividend payments and/or voting rights. Google has three classes of stock, with the class B stock having more voting rights than class A shares. These are

▼ **TABLE 14.6** NVIDIA Corporation and Subsidiaries Consolidated Balance Sheets (in millions, except par value)

	January 26, 2020	January 27, 2019
Assets		
Current assets:		
Cash and cash equivalents	\$10,896	\$ 782
Marketable securities	1	6,640
Accounts receivable, net	1,657	1,424
Inventories	979	1,575
Prepaid expenses and other current assets	<u>157</u>	<u>136</u>
Total current assets	13,690	10,557
Property and equipment, net	1,674	1,404
Operating lease assets	618	—
Goodwill	618	618
Intangible assets, net	49	45
Deferred income tax assets	548	560
Other assets	<u>118</u>	<u>108</u>
Total assets	<u>\$ 17,315</u>	<u>\$13,292</u>
Liabilities and shareholders' equity		
Current liabilities:		
Accounts payable	\$ 687	\$ 511
Accrued and other current liabilities	<u>1,097</u>	<u>818</u>
Total current liabilities	1,784	1,329
Long-term debt	1,991	1,988
Long-term operating lease liabilities	561	—
Other long-term liabilities	<u>775</u>	<u>633</u>
Total liabilities	5,111	3,950
Commitments and contingencies		
Shareholders' equity:		
Preferred stock, \$.001 par value; 2 shares authorized; none issued	—	—
Common stock, \$.001 par value; 2,000 shares authorized; 955 shares issued and 612 outstanding as of January 26, 2020; 945 shares issued and 606 outstanding as of January 27, 2019	1	1
Additional paid-in capital	7,045	6,051
Treasury stock, at cost (342 shares in 2020 and 339 shares in 2019)	(9,814)	(9,263)
Accumulated other comprehensive income (loss)	1	(12)
Retained earnings	<u>14,971</u>	<u>12,565</u>
Total shareholders' equity	<u>12,204</u>	<u>9,342</u>
Total liabilities and shareholders' equity	<u>\$ 17,315</u>	<u>\$13,292</u>

statement of cash flows explains how the company's cash changed from the beginning of the accounting period to the end.

sometimes called founder's shares and allow the founders to maintain control over the company even though they do not own the majority of the shares. Ford Motor has the same type of voting structure. Because each type of stock issued represents a different claim on the organization, each must be represented by a separate owners' equity account, called contributed capital.

The Statement of Cash Flows

The third primary financial statement is called the **statement of cash flows**, which explains how the company's cash changed from the beginning of the accounting period to the end. Cash, of course, is an asset shown on the balance sheet, which provides a snapshot of the firm's financial position at one point in time. However, many investors and other users of financial statements want more information about the cash flowing into and out of the firm than is provided on the balance sheet in order to better understand the company's financial health. The statement of cash flows takes the cash balance from one year's balance sheet and compares it with the next while providing detail about how the firm used the cash. Table 14.7 presents Nvidia statement of cash flows.

The change in cash is explained through details in three categories: cash from (used for) operating activities, cash from (used

for) financing activities, and cash from (used for) investing activities.

Cash from operating activities is calculated by combining the changes in the revenue accounts, expense accounts, current asset accounts, and current liability accounts. This category of cash flows includes all the accounts on the balance sheet that relate to computing revenues and expenses for the accounting period. If this amount is a positive number, as it is for Nvidia, then the business is making extra cash that it can use to invest in increased long-term capacity or to pay off debts such as loans or bonds.

Cash from investing activities is calculated from changes in the long-term or fixed asset accounts. If this amount is negative, as is the case with Nvidia, we can see that the company bought \$489 million of property and equipment. It also purchased \$1.461 billion of marketable securities and sold \$3.365 billion of investments and had \$4.744 billion of investments mature for a total positive cash flow of \$6.145 billion.

Cash from financing activities is calculated from changes in the long-term liability accounts and the contributed capital accounts in owners' equity. If this amount is negative, the company is likely paying off long-term debt or returning contributed capital to investors. In the case of Nvidia, it paid dividends of \$390 million, received \$149 million from employees purchasing stock through employee plans, and it paid taxes of \$551 million on restricted stock distributed to employees.

▼ **TABLE 14.7** NVIDIA Corporation and Subsidiaries Consolidated Statements of Cash Flows (in millions)

	Year Ended		
	January 26, 20 20	January 27, 2019	January 28, 2018
Cash flows from operating activities:			
Net income	\$2,796	\$4,141	\$3,047
Adjustments to reconcile net income to net cash provided by operating activities:			
Stock-based compensation expense	844	557	391
Depreciation and amortization	381	262	199
Deferred income taxes	18	(315)	(359)
Loss on early debt conversions	—	—	19
Other	5	(45)	20
Changes in operating assets and liabilities:			
Accounts receivable	(233)	(149)	(440)
Inventories	597	(776)	—
Prepaid expenses and other assets	77	(55)	21
Accounts payable	194	(135)	90
Accrued and other current liabilities	54	256	33
Other long-term liabilities	28	2	481
Net cash provided by operating activities	4,761	3,743	3,502

(Continued)

▼ **TABLE 14.7** (Continued)

	Year Ended		
	January 26, 2020	January 27, 2019	January 28, 2018
Cash flows from investing activities:			
Proceeds from maturities of marketable securities	4,744	7,232	1,078
Proceeds from sales of marketable securities	3,365	428	863
Purchases of marketable securities	(1,461)	(11,148)	(36)
Purchases of property and equipment and intangible assets	(489)	(600)	(593)
Investments and other, net	(14)	(9)	(36)
Proceeds from sale of long-lived assets and investments	—	—	2
Net cash provided by (used in) investing activities	<u>6,145</u>	<u>(4,097)</u>	<u>1,278</u>
Cash flows from financing activities:			
Payments related to repurchases of common stock	—	(1,579)	(909)
Repayment of Convertible Notes	—	(16)	(812)
Dividends paid	(390)	(371)	(341)
Proceeds related to employee stock plans	149	137	139
Payments related to tax on restricted stock units	(551)	(1,032)	(612)
Other	—	(5)	(9)
Net cash used in financing activities	<u>(792)</u>	<u>(2,866)</u>	<u>(2,544)</u>
Change in cash and cash equivalents	10,114	(3,220)	2,236
Cash and cash equivalents at beginning of period	<u>782</u>	<u>4,002</u>	<u>1,766</u>
Cash and cash equivalents at end of period	<u>\$10,896</u>	<u>\$ 782</u>	<u>\$4,002</u>

LO 14-5 Analyze financial statements, using ratio analysis, to evaluate a company's performance.

RATIO ANALYSIS: ANALYZING FINANCIAL STATEMENTS

The income statement shows a company's profit or loss, while the balance sheet itemizes the value of its assets, liabilities, and owners' equity. Together, the two statements provide the means to answer two critical questions: (1) How much did the firm make or lose? and (2) How much is the firm presently worth based on historical values found on the balance sheet? **Ratio analysis**, calculations that measure an organization's financial health, brings the complex information from the income statement and balance sheet into sharper focus so that managers, lenders, owners, and other interested parties can measure and compare the organization's productivity, profitability, and

financing mix with other similar entities.

As you know, a ratio is simply one number divided by another, with the result showing the relationship between the two numbers. For example, we measure fuel efficiency with miles per gallon. This is how we know that 55 mpg in a Toyota Prius is much better than the average car. We use ratios in all sports, such as earned run and batting averages in baseball, field goal percentage in basketball, and percentage of passes completed in football. But, to make sense out of ratios, you have to know what you want to measure. Financial ratios are used to weigh and evaluate a firm's performance. An absolute value such as earnings of \$70,000 or accounts receivable of \$200,000 almost never provides as much useful information as a well-constructed ratio. Whether those numbers are good or bad depends on their relation to other numbers. If a company earned \$70,000 on \$700,000 in sales (a 10 percent return), such an earnings level might be quite satisfactory. The president of a company earning this same \$70,000 on sales of \$7 million (a 1 percent return), however, should probably start looking for another job!

ratio analysis

calculations that measure an organization's financial health.

profitability ratios

ratios that measure the amount of operating income or net income an organization is able to generate relative to its assets, owners' equity, and sales.

profit margin net income divided by sales.

return on assets net income divided by assets.

return on equity net income divided by owners' equity; also called return on investment (ROI).

asset utilization ratios ratios that measure how well a firm uses its assets to generate each \$1 of sales.

receivables turnover sales divided by accounts receivable.

inventory turnover sales divided by total inventory.

Ratios by themselves are not very useful. What matters is the relationship of the calculated ratios to the previous years performance, comparison to its competitors and the company's stated goals. Remember, while the profitability, asset utilization, liquidity, debt ratios, and per share data we'll look at here can be very useful, you will never see the forest by looking only at the trees.

Profitability Ratios

Profitability ratios measure how much operating income or net income an organization is able to generate relative to its assets, owners' equity, and sales. The numerator (top number) used in these examples is always the net income after taxes. Common profitability ratios include profit margin, return on assets, and return on equity. The following examples are based on the 2020 income statement and balance sheet for Nvidia, as shown in Tables 14.4 and 14.6. Except where specified, all data are expressed in millions of dollars.

The **profit margin**, computed by dividing net income by sales, shows the overall percentage of profits earned by the company. It is based solely upon data obtained from the income statement. The higher the profit margin, the better the cost controls within the company and the higher the return on every dollar of revenue. Nvidia's profit margin is calculated as follows:

$$\begin{aligned}\text{Profit margin} &= \frac{\text{Net income}}{\text{Sales}} \\ &= \frac{\$2,796}{\$10,918} \\ &= 25.61\%\end{aligned}$$

Thus, for every \$1 in sales, Nvidia generated profits after taxes of slightly more than 25 cents.

Return on assets, net income divided by assets, shows how much income the firm produces for every dollar invested in assets. A company with a low return on assets is probably not using its assets very productively—a key managerial failing. For

its computation, the return on assets calculation requires data from both the income statement and the balance sheet.

$$\begin{aligned}\text{Return on assets} &= \frac{\text{Net income}}{\text{Total assets}} \\ &= \frac{\$2,796}{\$17,315} \\ &= 16.15\%\end{aligned}$$

In the case of Nvidia, every \$1 of assets generated a return of 16.15 percent, or profits of 16.15 cents per dollar.

Stockholders are always concerned with how much money they will make on their investment, and they frequently use the return on equity ratio as one of their key performance yardsticks.

Return on equity (also called return on investment [ROI]), calculated by dividing net income by owners' equity, shows how much income is generated by each \$1 the owners have invested in the firm. Obviously, a low return on equity means low stockholder returns and may indicate a need for immediate managerial attention. Because some assets may have been financed with debt not contributed by the owners, the value of the owners' equity is usually considerably lower than the total value of the firm's assets. Nvidia's return on equity is calculated as follows:

$$\begin{aligned}\text{Return on equity} &= \frac{\text{Net income}}{\text{Stockholders' equity}} \\ &= \frac{\$2,796}{\$12,204} \\ &= 22.91\%\end{aligned}$$

For every dollar invested by Nvidia stockholders, the company earned a 22.91 percent return, or 22.91 cents per dollar invested.

Asset Utilization Ratios

Asset utilization ratios measure how well a firm uses its assets to generate each \$1 of sales. Obviously, companies using their assets more productively will have higher returns on assets than their less efficient competitors. Similarly, managers can use asset utilization ratios to pinpoint areas of inefficiency in their operations. These ratios (receivables turnover, inventory turnover, and total asset turnover) relate balance sheet assets to sales, which are found on the income statement.

The **receivables turnover**, sales divided by accounts receivable, indicates how many times a firm collects its accounts receivable in one year. It also demonstrates how quickly a firm is able to collect payments on its credit sales. Obviously, no payments means no profits. Nvidia collected its receivables 6.59 times per year, which translates

“Stockholders are always concerned with how much money they will make on their investment, and they frequently use the return on equity ratio as one of their key performance yardsticks.”

total asset**turnover** sales divided by total assets.**liquidity ratios**

ratios that measure the speed with which a company can turn its assets into cash to meet short-term debt.

current ratio

current assets divided by current liabilities.

quick ratio (acid**test)** a stringent measure of liquidity that eliminates inventory.**debt utilization****ratios** ratios that measure how much debt an organization is using relative to other sources of capital, such as owners' equity.

to about 55 days that receivables are outstanding. These ratios are based on a 360 day year. This is most likely due to the trade terms it gives its corporate customers.

$$\begin{aligned}\text{Receivables turnover} &= \frac{\text{Sales}}{\text{Receivables}} \\ &= \frac{\$10,918}{\$1,657} \\ &= 6.59 \text{ times per year}\end{aligned}$$

Inventory turnover, sales divided by total inventory, indicates how many times a firm sells and replaces its inventory over the course of a year. A high inventory turnover ratio may indicate great efficiency but may also suggest the possibility of lost sales due to insufficient stock levels. Nvidia's inventory turnover indicates that it replaced its inventory 11.15 times last year, or about every 32 days.

$$\begin{aligned}\text{Inventory turnover} &= \frac{\text{Sales}}{\text{Inventory}} \\ &= \frac{\$10,918}{\$979} \\ &= 11.15 \text{ times per year}\end{aligned}$$

Accountants often like to calculate inventory turnover using cost of goods sold in the numerator because inventory is carried on the books at cost and so is cost of goods sold. Financial analysts prefer sales in the numerator to measure the efficiency of inventory in producing revenues. Nvidia makes very expensive high-end computer chips that power video games and powerful servers used in cloud operations. Their inventory turnover is quite good.

Total asset turnover, sales divided by total assets, measures how well an organization uses all of its assets in creating sales. It indicates whether a company is using its assets productively. Nvidia generated \$0.63 in sales for every \$1 in total corporate assets. The cause of this low total asset turnover is the large cash balance that Nvidia has on its balance sheet. Cash does not produce sales income. At the end of the fiscal year more than 62 percent of its assets were held in cash or cash equivalents.

$$\begin{aligned}\text{Total asset turnover} &= \frac{\text{Sales}}{\text{Total assets}} \\ &= \frac{\$10,918}{\$17,315} \\ &= 0.63 \text{ times per year}\end{aligned}$$

Liquidity Ratios

Liquidity ratios compare current (short-term) assets to current liabilities to indicate the speed with which a company can turn its assets into cash to meet debts as they fall due. High liquidity ratios may satisfy a creditor's need for safety, but ratios that are too high may indicate that the organization is not using its current assets efficiently. Liquidity ratios are generally best examined in conjunction with asset utilization ratios because high turnover ratios imply that cash is flowing through an organization very quickly—a situation that dramatically reduces the need for the type of reserves measured by liquidity ratios.

The **current ratio** is calculated by dividing current assets by current liabilities. Nvidia's current ratio indicates that for every \$1 of current liabilities, the firm had \$7.67 of current assets on hand. This very high current ratio is also due to the \$10.896 billion of cash, and cash equivalents, on hand, which is part of the current asset total.

$$\begin{aligned}\text{Current ratio} &= \frac{\text{Current assets}}{\text{Current liabilities}} \\ &= \frac{\$13,690}{\$1,784} \\ &= 7.67 \text{ times}\end{aligned}$$

The **quick ratio** (also known as the **acid test**) is a far more stringent measure of liquidity because it eliminates inventory, the least liquid current asset. It measures how well an organization can meet its current obligations without resorting to the sale of its inventory. Because Nvidia carries a small amount of inventory, the quick ratio at 7.13 times is almost the same as the current ratio.

$$\begin{aligned}\text{Quick ratio} &= \frac{\text{Current assets} - \text{Inventory}}{\text{Current liabilities}} \\ &= \frac{\$13,690 - \$979}{\$1,784} \\ &= 7.13 \text{ times}\end{aligned}$$

Debt Utilization Ratios

Debt utilization ratios provide information about how much debt an organization is using relative to other sources of capital, such as owners' equity. Because the use of debt carries an interest charge that must be paid regularly regardless of profitability, debt financing is much riskier than equity. Unforeseen negative events such as recessions affect heavily indebted firms to a far greater extent than those financed exclusively with

debt to total assets

ratio a ratio indicating how much of the firm is financed by debt and how much by owners' equity.

times interest

earned ratio operating income divided by interest expense.

per share data

data used by investors to compare the performance of one company with another on an equal, per share, basis.

earnings per

share net income or profit divided by the number of stock shares outstanding.

dividends per

share the actual cash received for each share owned.

owners' equity. Because of this and other factors, the managers of most firms tend to keep debt-to-asset levels below 50 percent. However, firms in very stable and/or regulated industries, such as electric utilities, often are able to carry debt ratios well in excess of 50 percent with no ill effects.

The **debt to total assets ratio** indicates how much of the firm is financed by debt and how much by owners' equity. To find the value of Nvidia's total debt, you must add current liabilities to long-term debt and other liabilities or simply subtract stockholders' equity from total assets.

$$\begin{aligned}\text{Debt to total assets} &= \frac{\text{Debt (Assets} - \text{Equity)}}{\text{Total assets}} \\ &= \frac{\$17,315 - \$12,204}{\$17,315} \\ &= 30\%\end{aligned}$$

Thus, for every \$1 of Nvidia's total assets, 30 percent is financed with debt. The remaining 70 percent is provided by owners' equity.

The **times interest earned ratio**, operating income divided by interest expense, is a measure of the safety margin a company has with respect to the interest payments it must make to its creditors. A low times interest earned ratio indicates that even a small decrease in earnings may lead the company into financial straits. We find that interest expense on the income statement was 52 million and that it has gone down by a small amount over the last several years. Putting this into the calculation, we find that interest expense is covered 54.73 times by operating income. This is a very high number which shows that Nvidia would have no problem paying its interest on its debt. In fact lenders would be happy to loan Nvidia money.

$$\begin{aligned}\text{Times interest earned} &= \frac{\text{Income before interest and taxes}}{\text{Interest}} \\ &= \frac{\$2,846}{\$52} \\ &= 54.73 \text{ times}\end{aligned}$$

Per Share Data

Investors may use **per share data** to compare the performance of one company with another on an equal, or per share, basis. Generally, the more shares of stock a company issues, the less income is available for each share.

Earnings per share is calculated by dividing net income or profit by the number of shares of stock outstanding. This ratio is important because yearly changes in earnings per share, in combination with other economy-wide factors, determine

a company's overall stock price. When earnings go up, so does a company's stock price—and so does the wealth of its stockholders.

$$\begin{aligned}\text{Diluted earnings} &= \frac{\text{Net income}}{\text{Number of shares outstanding (diluted)}} \\ \text{per share} &= \frac{\$2,796}{618} \\ &= \$4.52\end{aligned}$$

We can see from the income statement that Nvidia's earnings per share are up and down between 2018 and 2020, and this is because the computer chip industry is somewhat cyclical as newer and faster chips come into the market every several years. So, rather than a steady increase year after year, investors can expect big ups and downs in earnings. You can see from the income statement that diluted earnings per share include more shares than the basic calculation; this is because diluted shares include potential shares that could be issued due to the exercise of stock options or the conversion of certain types of debt into common stock. Investors generally pay more attention to diluted earnings per share than basic earnings per share.

Dividends per share are paid by the corporation to the stockholders for each share owned. The payment is made from earnings after taxes by the corporation but is taxable income to the stockholder. Thus, dividends result in double taxation: The corporation pays tax once on its earnings, and the stockholder pays tax a second time on his or her dividend income. Nvidia—with almost \$11 billion in revenue—is a young, growing company compared to Intel with approximately \$75 billion in revenue. It only pays a small dividend, and as you can see on the income statement it spends a lot on research and development. Note that the shares listed on the income statement versus shares listed as outstanding on the balance sheet are not the same. Share count for earnings per share are weighted average shares over the year. The shares on the balance sheet are those outstanding at the year's end and do not necessarily represent the shares on which dividends were actually paid. Nvidia does not show dividends on the income statement and so you have to look on the statement of cash flows for total dividends paid.

$$\begin{aligned}\text{Dividends} &= \frac{\text{Dividends paid}}{\text{Number of shares outstanding (year end)}} \\ \text{per share} &= \frac{\$390}{609} \\ &= \$0.64\end{aligned}$$

WOMEN IN ACCOUNTING: GENDER EQUALITY ISN'T ADDING UP

Though three of the four Big Four accounting firms have women in high-level positions, a large gap exists between the number of men and women who make it to partner or principal. Approximately 61 percent of accountants and auditors in the United States are women, and 51 percent of all full-time staff at CPA firms are women. Women make up 47 percent of senior managers but only 24 percent of partners and principals.

Statistics also show that women are extremely underrepresented among accounting-firm partners who head audit engagement of some U.S. companies. Out of the S&P 100 companies, only 11 percent of the engagement partners are women. Engagement partners are the

lead representatives for client engagements. These numbers show that women are not being assigned to as many large clients as men.

One former Big Four accountant claims the discrepancy might not be due to the firms themselves. Instead, many women quit before making it to partner. Big Four firms are known to have very competitive cultures, requiring long hours. Accounting firm leaders have started to address these challenges by looking into creating a more family-friendly environment so accountants can take advantage of opportunities and achieve work/life balance. If successful, these changes may lead to more women taking on a partner or principal role in the accounting field—not only increasing diversity in leadership

but also possibly decreasing the high turnover rates at these firms.^d

Critical Thinking Questions

1. What are some reasons women might not be as represented among the partners or principals of the Big Four accounting firms?
2. Describe the corporate culture at the Big Four accounting firms. What might be done to make it more family friendly?
3. Do you think accounting firms should work to create a better work/life balance for employees with families, or should accountants expect to work long hours and make sacrifices if they choose to work for the Big Four?

IMPORTANCE OF INTEGRITY IN ACCOUNTING

The financial crisis and the recession that followed provided another example of a failure in accounting reporting. Many firms attempted to exploit loopholes and manipulate accounting processes and statements. Banks and other financial institutions often held assets off their books by manipulating their accounts. If the accountants, the SEC, and the bank regulators had been more careful, these types of transactions would have been discovered and corrected.

On the other hand, strong compliance to accounting principles creates trust among stakeholders. Accounting and financial planning is important for all organizational entities, even cities. The City of Maricopa in Arizona received the Government Finance Officers Association of the United States and Canada (GFOA) Distinguished Budget Presentation Award for its governmental budgeting. The city scored proficient in its policy, financial plan, operations guide, and communications device. Integrity in accounting is crucial to creating trust, understanding the financial position of an organization or entity, and making financial decisions that will benefit the organization.⁵

BUILDING YOUR SOFT SKILLS

BY THINKING ABOUT ETHICS

As an employee and even as a student, you may have to turn in expense reports for reimbursement. Approximately 14 percent of all fraud relates to expense report fraud. Some employees charge items that are for personal use, and others get blank receipts and fill in an amount greater than the expense. Think about your own personal ethics and explain how you will handle expense reimbursements. What if your colleagues tell you to add an extra amount because you probably forgot to include some other expenses?

Team Exercise

You can look at websites such as Yahoo! Finance (<http://finance.yahoo.com/>), under the company's "key statistics" link, to find many of its financial ratios, such as return on assets and return on equity. Have each member of your team look up a different company, and explain why you think there are differences in the ratio analysis for these two ratios among the selected companies.

It is most important to remember that integrity in accounting processes requires ethical principles and compliance with both the spirit of the law and professional standards in the accounting profession. Most states require accountants preparing to take the CPA exam to take accounting ethics courses. Transparency and accuracy in reporting revenue, income, and assets develops trust from investors and other stakeholders.

WOULD YOU MAKE // a Good Accountant? /



Do you like numbers and finances? Are you detail oriented, a perfectionist, and highly accountable for your decisions? If so, accounting may be a good field for you. If you are interested in accounting, there are always job opportunities available no matter the state of the economy. Accounting is one of the most secure job options in business. Of course, becoming an accountant is not easy. You will need at least a bachelor's degree in accounting to get a job, and many positions require additional training. Many states demand coursework beyond the 120 to 150 credit hours collegiate programs require for an accounting degree. If you are really serious about getting into the accounting field, you will probably want to consider getting your master's in accounting and taking the CPA exam. The field of accounting can be complicated, and the extra training provided through a master's in accounting program

will prove invaluable when you go out looking for a good job. Accounting is a volatile discipline affected by changes in legislative initiatives.

With corporate accounting policies changing constantly and becoming more complex, accountants are needed to help keep a business running smoothly and within the bounds of the law. In fact, the number of jobs in the accounting and auditing field are expected to increase 10 percent between 2016 and 2026. Jobs in accounting tend to pay quite well, with the median salary standing at \$69,350. If you go on to get your master's degree in accounting, expect to see an even higher starting wage. Of course, your earnings could be higher or lower than these averages, depending on where you work, your level of experience, the firm, and your particular position.

Accountants are needed in the public and the private sectors, in large and small firms,

in for-profit and not-for-profit organizations. Accountants in firms are generally in charge of preparing and filing tax forms and financial reports. Public-sector accountants are responsible for checking the veracity of corporate and personal records in order to prepare tax filings. Basically, any organization that has to deal with money and/ or taxes in some way or another will be in need of an accountant, either for in-house service or occasional contract work. Requirements for audits under the Sarbanes-Oxley Act and rules from the Public Company Accounting Oversight Board are creating more jobs and increased responsibility to maintain internal controls and accounting ethics. The fact that accounting rules and tax filings tend to be complex virtually ensures that the demand for accountants will never decrease.⁶ ■